

From Einstein-Boltzmann Equations to Galaxy Power Spectra: Physicist-Supervised, Oracle-Guided AI Reimplementation in JAX

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Abstract

Gradient-based inference methods such as Hamiltonian Monte Carlo and simulation-based inference require differentiable forward models, yet no existing code provides full CMB spectra plus one-loop galaxy power spectra with exact automatic differentiation on GPU. We address this gap with CLAX and CLAX-PT, fully differentiable JAX reimplementations of the CLASS Boltzmann solver and its one-loop perturbation theory extension CLASS-PT, developed via oracle-guided agentic development: AI agents iterate against reference data from CLASS and CLASS-PT, autonomously resolving implementation bugs and escalating to human review only when oracle feedback stalls. Across 57 Claude Code sessions spanning 12 days, agents autonomously resolved 38 of 40 documented bugs (95%) without human input. The remaining 5% required human physics judgment and fell into exactly two failure modes: *architectural incoherence* (code structure incompatible with the target physics) and *oracle-passing fudge factors* (a numerical patch satisfying all tests but encoding no physical model). CLAX achieves $< 0.1\%$ CMB accuracy at $\ell \leq 2000$; CLAX-PT achieves $< 0.7\%$ galaxy power spectrum accuracy for monopole and quadrupole spectra at $k < 0.3 \text{ h/Mpc}$. Automatic differentiation gradients agree with fourth-order finite differences to 0.15% on average. The full pipeline runs at 34 s/step on a single V100 GPU with HMC-ready accuracy. We characterize the boundary between unsupervised and supervised modes of AI-assisted scientific development, propose three principles for structuring human-agent collaboration, and compare against concurrent differentiable cosmology codes DISCO-EB and ABCMB.

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1. Introduction

The standard model of cosmology is tested by comparing theoretical predictions against observations from galaxy surveys such as DESI (DESI Collaboration, 2025), Euclid, and CMB experiments including Planck. This comparison increasingly demands gradient-based inference: exact Fisher matrices, HMC sampling (Neal, 2011), and simulation-based inference (Cranmer et al., 2020) all benefit from differentiable forward models. Libraries such as NumPyro (Phan et al., 2019), BlackJAX (Cabezas et al., 2024), and adaptive MCMC with normalizing flows (Gabri  et al., 2022) provide state-of-the-art gradient-based samplers, but the missing ingredient is a differentiable cosmological forward model that can be passed to them.

The dominant Boltzmann codes—CLASS (Blas et al., 2011) and CAMB (Lewis et al., 2000)—are written in C and Fortran, are not differentiable, and are not GPU-native. Several JAX implementations exist but cover only subsets of the problem: DISCO-EB (Hahn et al., 2023) computes the nonlinear matter power spectrum $P(k)$ but not CMB spectra or galaxy clustering observables; ABCMB (Zhou et al., 2026) implements the CMB pipeline including lensing and massive neutrinos but does not include perturbation theory for galaxy surveys; JAX-COSMO (Lemos et al., 2023) computes the nonlinear matter power spectrum but not the full Boltzmann hierarchy; and SymBoltz.jl (Sletmoen, 2024) provides full CMB spectra in Julia but without GPU support. No existing code provides the full chain from cosmological parameters through CMB angular power spectra and one-loop EFT galaxy power spectrum multipoles, with end-to-end automatic differentiation on GPU.

We describe the development of CLAX and CLAX-PT—fully differentiable JAX reimplementations of the CLASS and CLASS-PT pipelines—and the oracle-guided agentic methodology used to build them. Using CLASS v3.3.4 and CLASS-PT (Chudaykin et al., 2021) as ground truth, we find a clear boundary: agents handle implementation bugs (convention errors, algorithmic transcription, coefficient verification) autonomously and reliably, but fail at two classes of problem that require human physics judgment: (1) recognizing that an architecture is structurally incompatible with the intended physics, and (2) detecting that a numerically ef-

fective patch satisfies all oracle tests at the calibration point while encoding incorrect physics.

Contributions.

- CLAX: a JAX reimplementation of the full CLASS CMB pipeline (background, thermodynamics, perturbations, transfer, lensing), achieving $< 0.1\%$ accuracy at $\ell \leq 2000$ with JIT-compiled GPU execution at 34 s/step (V100, `fit_cl` preset).
- CLAX-PT: a JAX reimplementation of CLASS-PT’s one-loop EFT power spectra with IR resummation and RSD multipoles, achieving $< 0.7\%$ accuracy for monopole and quadrupole spectra.
- Validated autodifferentiation gradients matching finite differences to 0.15% across cosmological parameters.
- A landscape comparison against DISCO-EB, ABCMB, CLASS, and CLASS-PT, showing that CLAX is the first code combining CMB and one-loop galaxy spectra with full AD on GPU.
- An empirical bug taxonomy of 40 bugs across the development, classifying autonomous vs. human-required resolution and identifying two specific failure modes of oracle-guided development.

2. Background

2.1. Cosmological inference and differentiable Boltzmann codes

Galaxy surveys and CMB experiments measure the angular power spectrum C_ℓ and the galaxy power spectrum $P_{\text{gg}}(k, \ell)$, which depend on cosmological parameters θ (density, dark matter density, Hubble constant, spectral index, etc.) through the Boltzmann equation—a coupled system of ODEs for photon, baryon, dark matter, and neutrino perturbations that must be integrated numerically. An exact Fisher matrix for parameter inference requires $\partial C_\ell / \partial \theta_i$; HMC requires the full gradient $\nabla_\theta \log P(\theta | \text{data})$; both are expensive to approximate by finite differences and exact only with automatic differentiation (AD). CLAX enables `jax.grad`(C_ℓ) (params) for the first time in the CMB context, and CLAX-PT extends this to one-loop galaxy power spectrum multipoles.

2.2. AI-assisted scientific code development

The oracle-based methodology we employ is inspired by the AI C compiler project (Carlini, 2026), in which 16 parallel Claude Code agents produced a 100k-line Rust C compiler over ≈ 2000 sessions, using GCC as an oracle. Key infrastructure includes: (i) a shared `CHANGELOG.md` updated

after every session to prevent re-exploration of dead ends; (ii) test-first development where every module’s tests are written against reference data before implementation; (iii) a `--fast` flag sampling 10% of test cases to prevent “time blindness”; and (iv) parallel agent teams using git worktrees to explore competing hypotheses independently.

The critical challenge not addressed in the compiler setting is *domain knowledge*: CLASS is not just a large codebase—it encodes decades of cosmological physics knowledge, conventions, and approximations that require physics expertise to interpret. This makes the boundary between supervised and unsupervised development especially important to characterize.

3. Methodology

3.1. Pipeline architecture

CLAX implements a sequential pipeline of pure functions, each returning a frozen PyTree:

$\text{CosmoParams} \rightarrow \text{BG} \rightarrow \text{TH} \rightarrow \text{PT} \rightarrow \text{PR} \rightarrow \text{TR} \rightarrow \text{HR} \rightarrow \text{LE}$

where BG is background cosmology, TH is thermodynamics/recombination, PT is perturbation theory (Einstein–Boltzmann hierarchy), PR is the primordial spectrum, TR is transfer functions, HR is harmonic (C_ℓ) integration, and LE is CMB lensing via Wigner d -functions. `CosmoParams` fields are JAX-traced (for AD); `PrecisionParams` fields are static (control array shapes). No branching on traced values ensures compatibility with `jax.jit` and `jax.grad`.

The Einstein–Boltzmann system is integrated using Kvaerno5 (ESDIRK, default) or Rodas5 (Rosenbrock) solvers from the `DiffraX` library (Kidger, 2021), with a DISCO-EB-style filtered PID step controller using k -dependent weighting on six key perturbation variables. Reverse-mode AD through the ODE solve uses `RecursiveCheckpointAdjoint` for memory efficiency, with `custom_vjp` wrappers for the shooting method ($\theta_s \rightarrow H_0$ via implicit differentiation) and the Saha equilibrium (implicit-function-theorem JVP).

CLAX-PT extends the pipeline with one-loop EFT power spectra following the CLASS-PT algorithm: FFTLog decomposition of P_{22} and P_{13} integrals (Fang et al., 2017), precomputed M_{22}/M_{13} kernel matrices, IR resummation via DST with odd/even spline mode removal (Senatore & Zaldarriaga, 2015; Vlah et al., 2015), an EFT bias expansion ($b_1, b_2, b_{s^2}, b_{3\text{nl}}$, counterterms, stochastic contributions), and RSD multipoles ($\ell = 0, 2, 4$) via Gauss–Legendre quadrature over μ with anisotropic BAO damping. The entire CLAX-PT module ($\sim 2,100$ lines) is end-to-end differentiable: `jax.grad` flows through FFTLog, IR resummation, bias assembly, and GL quadrature.

3.2. Development methodology

CLAX was developed primarily by the first author over ~ 6 days using Claude Code (Opus 4.6), producing $\sim 10,000$ lines of JAX code across ≥ 100 agent sessions. CLAX-PT was developed by the second author over 12 days and 57 logged sessions, producing $\sim 2,100$ lines. Both phases used the same oracle-guided methodology: every module has a test file written *before* implementation, specifying what CLASS (or CLASS-PT) produces; the code is finished when and only when the oracle tests pass.

The supervision methodology—encoded in `CLAUDE.md` and refined across both development phases—proved essential. Two rules were particularly important: (1) never add fudge factors (if a test fails at 0.2%, find the actual bug), and (2) test at multiple parameter points, not just the fiducial cosmology. Both rules directly prevented specific failure modes documented in Section 5.

4. Results: Accuracy and Performance

4.1. Landscape comparison

Table 1 compares CLAX against existing differentiable and non-differentiable cosmology codes. CLAX is the first JAX code providing both CMB angular power spectra and one-loop galaxy power spectrum multipoles with end-to-end automatic differentiation on GPU.

4.2. CMB pipeline accuracy

Table 2 summarizes accuracy of CLAX against CLASS at the Planck 2018 best-fit cosmology (`planck_cl` preset). Background quantities match to $< 0.01\%$; thermodynamics to $< 0.1\%$. CMB temperature and polarization spectra are within $0.02\%–0.57\%$ across $\ell = 20–1000$. Lensed spectra at $\ell = 2000$ pass at $< 0.2\%$. Multi-cosmology validation across 10 parameter points ($\omega_b/\text{cdm} \pm 20\%$, $h \pm 10\%$, massive neutrinos $0.06–0.3$ eV, $N_{\text{eff}} = 2.0–4.0$) confirms sub-0.5% accuracy throughout parameter space.

4.3. Galaxy power spectrum accuracy

Table 3 shows accuracy for one-loop EFT galaxy power spectra at $z = 0.38$ (BOSS effective redshift) against CLASS-PT. Real-space spectra (P_{mm} , P_{gg} , P_{gm}) agree to $< 0.31\%$ (mean 0.04%). RSD monopoles ($\ell = 0$) agree to $< 0.6\%$ for matter and galaxies. Quadrupoles ($\ell = 2$) agree to $< 0.7\%$ for matter and $< 1.1\%$ for galaxies (the galaxy quadrupole has a known residual error in a tree-level bias term). Hexadecapoles ($\ell = 4$) are validated using $|\Delta|/\max(|\text{ref}|) < 2\%$ since CLASS-PT values are small and the spectrum crosses near zero.

4.4. Gradient verification

Automatic differentiation gradients (reverse-mode, using `jax.grad`) agree with fourth-order centered finite differences to 0.15% on average, with 97.7% of k -modes passing a 1% threshold. Table 4 shows the parameter-by-parameter breakdown for the matter power spectrum $P_{mm}(k)$ at $k = 0.1 h/\text{Mpc}$.

Key technical ingredients enabling stable AD through the full pipeline include: (1) `RecursiveCheckpointAdjoint` for memory-efficient reverse-mode through the stiff perturbation ODE; (2) a `custom_vjp` on the shooting method ($\theta_s \rightarrow H_0$) implementing implicit differentiation; and (3) a `custom_jvp` on the hydrogen Saha equilibrium equation, replacing explicit `stop_gradient` calls with implicit-function-theorem tangents to restore nonzero sensitivities around recombination.

4.5. Performance

CLAX offers two presets for different use cases (Table 5). The `fit_cl` preset uses 20 k/decade , $\ell_{\text{max}} = 17$ hierarchy, table-based Bessel functions, and `rtol` = 10^{-3} to achieve 34 s/step on a V100 GPU—sufficient for HMC. The `planck_cl` preset uses 60 k/decade , $\ell_{\text{max}} = 50$, and tight tolerances for science-grade accuracy at ~ 487 s on H100. An alternative Rodas5 Rosenbrock solver (Di Marzo, 1993), following DISCO-EB, avoids Newton iteration and offers $\sim 3–5\times$ speedup on the perturbation ODE.

4.6. Application: gradient scaling

The primary advantage of AD over finite differences for cosmological inference is the scaling of gradient cost with the number of parameters d . Finite-difference gradients require $2d$ forward evaluations (centered differences); AD requires a single forward pass plus one backward pass regardless of d . For CLAX with $d = 6$ standard ΛCDM parameters, this represents a $\sim 12\times$ reduction in gradient cost. For extended models ($d = 10–20$ including neutrino masses, dark energy, bias parameters, and counterterms), the advantage grows to $20–40\times$. Combined with the 34 s forward evaluation of `fit_cl`, this makes gradient-based sampling with exact theory predictions feasible for the first time in full-shape galaxy clustering analysis.

5. Supervised vs. Unsupervised Development

5.1. Bug taxonomy

Over the full development of CLAX and CLAX-PT, we documented 40 bugs: 26 in the CMB pipeline and 14 in the PT module. We classify these into four types by their nature and whether autonomous resolution was possible.

Table 1. Comparison of cosmological theory codes. “AD” = automatic differentiation. “PT” = one-loop perturbation theory for galaxy clustering. CLAX is the only code combining CMB C_ℓ , lensing, $P(k)$, and 1-loop PT with full AD on GPU.

Code	Framework	C_ℓ	Lensing	$P(k)$	1-loop PT	AD	GPU
CLASS + CLASS-PT	C / Python	✓	✓	✓	✓	—	—
CAMB	Fortran	✓	✓	✓	—	—	—
DISCO-EB	JAX	—	—	✓	—	✓	✓
ABCMB	JAX	✓	✓	✓	—	✓	✓
SymBoltz.jl	Julia	✓	—	✓	—	✓	—
CLAX + CLAX-PT	JAX	✓	✓	✓	✓	✓	✓

Table 2. CMB accuracy (CLAX vs. CLASS, `planck_cl` preset, Planck 2018 fiducial).

Quantity	Max error
$H(z)$, $D_A(z)$, r_s	$< 0.01\%$
$x_e(z)$, $g(\tau)$	$< 0.1\%$
C_ℓ^{TT} , $\ell = 20, 100, 500, 1000$	0.08%, 0.02%, 0.15%, 0.57%
C_ℓ^{EE} , $\ell = 20, 100, 500, 1000$	0.21%, 0.02%, 0.15%, 0.26%
Lensed C_ℓ , $\ell = 2000$	$< 0.2\%$
Multi-cosmology (10 pts)	$< 0.5\%$

Table 3. Galaxy power spectrum accuracy (CLAX-PT vs. CLASS-PT, $z = 0.38$).

Spectrum	Max error	Metric
$P_{mm}^{\text{real}}, P_{gg}^{\text{real}}, P_{gm}^{\text{real}}$	0.31%	rel.
$P_{mm}^{(\ell=0)}, P_{gg}^{(\ell=0)}$	0.59%, 0.56%	rel.
$P_{mm}^{(\ell=2)}, P_{gg}^{(\ell=2)}$	0.70%, 0.89%	rel.
$P_{mm}^{(\ell=4)}, P_{gg}^{(\ell=4)}$	0.70%, 1.43%	abs/max

Type I: Convention and unit bugs. Errors in translating CLASS conventions into JAX: wrong sign, missing numerical factor, incorrect unit conversion. *Agents resolved all Type I bugs autonomously* by reading the corresponding CLASS source code and correcting the factor.

Type II: Algorithm implementation bugs. Errors in the numerical algorithm itself: wrong grid type, incorrect matrix storage format, incomplete angular kernel functions. *Agents resolved all Type II bugs autonomously*, typically by careful comparison of intermediate arrays against CLASS/CLASS-PT at specific (k, z) points.

Type III: Architectural bugs. The computational structure assumed by the implementation is fundamentally incapable of representing the correct physics. One instance occurred: the analytic Legendre projection architecture assumed isotropic BAO damping, making it unable to handle anisotropic damping $\Sigma_{\text{tot}}^2(\mu)$. *This required human intervention.*

Type IV: Physics judgment failures. A numerical patch satisfies all oracle tests but encodes physically unmotivated behavior. One instance occurred: a scalar fudge factor $\alpha = 0.27$ passed all fiducial tests but has no physical derivation.

Table 4. AD vs. finite-difference gradients for $P_{mm}(k = 0.1 \text{ h/Mpc})$.

Parameter	$ \text{AD}/\text{FD} - 1 $	Status
$\ln(10^{10} A_s)$	$< 0.01\%$	✓
n_s	$< 0.01\%$	✓
h	$< 0.5\%$	✓
ω_b	$< 1.0\%$	✓
ω_{cdm}	$< 0.5\%$	✓

Table 5. Performance summary.

Preset	GPU	Time	C_ℓ accuracy
<code>fit_cl</code>	V100-32GB	34 s	$< 1.5\%$ ($\ell \leq 500$)
<code>planck_cl</code>	H100-80GB	487 s	$< 0.2\%$ ($\ell \leq 2000$)

This required human intervention.

The resolution breakdown is: Types I+II (38 bugs): 100% autonomous; Types III+IV (2 bugs): 0% autonomous. Overall, 38/40 bugs (95%) were resolved without human input.

5.2. Principles for human–agent collaboration

We distill three principles from the empirical data:

P1: Oracle testing verifies *what*, not *why*. An oracle compares numerical values at a reference parameter point. It catches implementation errors (wrong sign, wrong coefficient) but not physics errors (wrong model, numerically compensated). A fudge factor with zero degrees of freedom at the calibration point will pass every oracle test at that point. *Defense*: test across diverse parameter space; test intermediate quantities; build auxiliary tests that check structural properties (e.g., that $P_{\text{tree}}(k, \mu)$ has the correct anisotropy as a function of μ), not just numerical values.

P2: Shared memory prevents re-exploration but not architectural loops. `CHANGELOG.md` records which algorithms were tried and failed, preventing agents from re-discovering dead ends. This was effective for Type I and II bugs. It did not prevent 33 sessions of exploration within the wrong architectural framework because all attempts were coherent within the (incorrect) model. *Defense*: when a bug persists after $N \approx 5\text{--}10$ sessions without progress, treat

the lack of progress itself as evidence of an architectural problem and trigger human review.

P3: The irreducible human role is architectural and physical judgment. Agents excel at reading CLASS source code and transcribing equations faithfully; identifying which term disagrees via targeted comparison with oracle outputs; and adjusting coefficients and conventions. Agents fail at recognizing that an architectural assumption is inconsistent with the physics, and at detecting that a numerically adequate patch produces zero test failures but is not physics. *Detecting physics errors that pass all tests requires knowledge of what the physics should look like*, which is a form of human supervision that oracle testing cannot replace.

6. Related Work

Differentiable cosmological codes. DISCO-EB (Hahn et al., 2023) implements the nonlinear matter power spectrum in JAX but does not compute CMB spectra (C_ℓ) or galaxy clustering observables. ABCMB (Zhou et al., 2026) provides a complete CMB pipeline in JAX including lensing, massive neutrinos, and a state-of-the-art HyReX recombination treatment, but does not include perturbation theory for galaxy surveys. SymBoltz.jl (Sletmoen, 2024) provides a full Boltzmann solver in Julia with automatic differentiation but without GPU support. JAX-COSMO (Lemos et al., 2023) computes the nonlinear matter power spectrum but not the full Boltzmann hierarchy and not CMB spectra. CLAX is the first JAX code to provide full CMB spectra plus one-loop galaxy power spectra with exact automatic differentiation and GPU execution.

EFT power spectra and IR resummation. The EFT of large-scale structure (Perko et al., 2016; D’Amico et al., 2020) provides a systematic expansion for the galaxy power spectrum. CLASS-PT (Chudaykin et al., 2021) is the standard implementation; CLAX-PT reproduces it in differentiable JAX. IR resummation handles BAO peak broadening from long-wavelength displacements (Senatore & Zaldarriaga, 2015; Vlah et al., 2015).

Agent-assisted scientific software. Recent work has shown that LLM agents can complete extended scientific tasks autonomously (Carlini, 2026), but most evaluations focus on syntactic correctness (compilation, test passage) rather than physical validity. Our work identifies semantic correctness—the requirement that a passing numerical patch has physical basis, not just calibration-point accuracy—as the key challenge distinguishing scientific software from general code generation.

7. Conclusion

We have developed CLAX and CLAX-PT—fully differentiable JAX reimplementations of the CLASS and CLASS-PT Boltzmann codes—using oracle-guided agentic development. The codes achieve $< 0.1\%$ CMB accuracy, $< 0.7\%$ galaxy power spectrum accuracy, with automatic differentiation gradients verified to 0.15% and GPU execution at 34 s/step on V100. Empirically, AI agents resolved 95% of implementation bugs autonomously. The remaining 5% required human physics judgment: recognizing a structural mismatch between an architecture and its target physics, and detecting a numerically effective but physically unmotivated fudge factor.

These findings suggest a practical framework for AI-assisted scientific code development: agents operate autonomously in unsupervised mode for implementation work, but human scientists remain in the loop as architectural and physical reviewers, intervening proactively when oracle feedback stalls or when a numerical patch lacks physical derivation. The codes are open source at <https://github.com/smsharma/clax>.

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